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Single Lead Electrocardiographic Detection of Left Ventricular Systolic Dysfunction in Pediatric and Congenital Heart Disease

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ABSTRACT

Portable, scalable, and accessible artificial intelligence (AI)-enabled smartwatch technology shows promise as a cardiovascular risk stratification strategy in the general adult population. The challenges of lifelong care in congenital heart disease (CHD)—inclusive of regional and socioeconomic disparities worldwide—underscore the need for similar solutions tailored to this population. Herein, we present the first noise-adapted single-lead ECG model for predicting left ventricular systolic dysfunction (left ventricular ejection fraction [LVEF] $\leq 40\%$) in pediatric and CHD patients. The internal cohort was comprised of 70,226 patients. External test groups included Children's Hospital of Philadelphia (CHOP; 42,984 patients) and Toronto General Hospital (TGH; 284 repaired tetralogy of Fallot patients). Our model had strong performance across a broad range of CHD lesions, races, ages, and healthcare systems. Our findings support the potential of AI-enabled wearables to expand global access to CHD care. Prospective studies utilizing wearable ECG devices in pediatric and CHD patients are warranted.

INTRODUCTION

Medical and surgical advancements have led to improved survival of children with congenital heart disease (CHD), with >90% reaching adulthood and >1 million adults with CHD in the United States.¹ This growing yet vulnerable population remains at increased risk of heart failure, a leading cause of death in CHD.² Left ventricular systolic dysfunction (LVSD) is independently associated with cardiovascular events in CHD, and guideline-directed medical therapy (GDMT) is associated with improvement in left ventricular ejection fraction (LVEF).³ Early detection of LVSD is therefore of considerable clinical interest, as it may facilitate timely initiation of disease-modifying therapies and consequent improvement in morbidity and mortality.

Accessible strategies for early detection of LVSD and heart failure in CHD remain limited. Clinical risk scores to predict heart failure in CHD—such as the adult CHD risk prediction model⁴—depend on clinical variables that require clinical evaluation, detailed medical histories, and laboratory testing (i.e., N-terminal pro-B type natriuretic peptide). These requirements inherently disadvantage individuals with limited access to healthcare or lapses in care, highlighting a critical need for a simplified and scalable method for LVSD and heart failure risk stratification in the CHD population.

As a foundational step toward addressing this gap, our group demonstrated that the inexpensive and ubiquitous 12-lead ECG contains hidden disease signatures of LVSD in pediatric and CHD that are detectable using artificial intelligence (AI) techniques.⁵ However, even the standard 12-lead ECG requires in-person clinical evaluation, and unfortunately nearly half of patients with CHD have gaps in care⁶ and are lost to follow-up.⁷ These trends are likely to be exacerbated by regionalization of care, sociodemographic factors, and disparate access to care worldwide.^{8,9} Moreover, the most vulnerable subsets of the CHD population (e.g., single

ventricle patients) have substantially higher rates of hemodynamic decompensation that necessitates close remote surveillance during the interstage period,¹⁰ and are similarly at risk of gaps in care over the lifespan.⁸

These considerations collectively underscore the need for a portable, scalable, and accessible risk stratification strategy tailored to the pediatric and CHD population. Portable single-lead ECG devices offer a promising alternative for AI-enabled remote cardiovascular surveillance in the general adult population.¹¹ However, similar technology has yet to be developed in the pediatric and CHD population in whom distinct epidemiological, anatomical, and electrophysiological characteristics require a dedicated pediatric and CHD algorithm.^{12,13}

In this study, we aim to address this technological gap by developing and externally validating an AI-enabled ECG (AI-ECG) model capable of analyzing noisy, single-lead ECG signals—representative of data collected from portable devices—to predict LVSD in a comprehensive cohort of pediatric and CHD patients (Figure 1).

RESULTS

Cohort Baseline Characteristics

The internal Boston Children's Hospital (BCH) training cohort comprised 133,344 ECG-echo pairs (52,670 patients; median age 10.7 [IQR, 3.5-16.8] years; 53% male), 6.8% with LVEF $\leq 50\%$ and 2.7% with LVEF $\leq 40\%$ (Table 1, Figure 1). Within this cohort, 2,579 patients had cardiomyopathy (11% with LVEF $\leq 40\%$), 2,027 had tetralogy of Fallot (1.9% with LVEF $\leq 40\%$), 249 had tricuspid atresia (5.7% with LVEF $\leq 40\%$), and 698 were ventricularly paced (13% with LVEF $\leq 40\%$). Similar lesion and outcome breakdowns were found in the internal testing cohort, which comprised of 45,151 ECG-echo pairs (17,556 patients; median age 10.5 [IQR, 3.5-16.8] years; 54% male), 6.9% with LVEF $\leq 50\%$ and 2.8% with LVEF $\leq 40\%$. Internal training and test cohort race/ethnicity demographics are shown in Table 1, with 59% White (non-Hispanic), 6% Black (non-Hispanic), 3% Asian (non-Hispanic), 9% Hispanic, 6-7% Other, and 16-17% Unknown. Approximately 24% of training and test patients had any form of CHD. For a comprehensive list of all individual CHD lesions analyzed at BCH, see Supplementary Table 1.

The Children's Hospital of Philadelphia (CHOP) external cohort comprised 76,400 ECG-echo pairs (42,984 patients; median age 10.8 [IQR, 4.9-15.0] years; 55% male), 3.8% with LVEF $\leq 50\%$ and 1.7% with LVEF $\leq 40\%$ (Table 1, Figure 1). Within this cohort, 1,808 patients had cardiomyopathy (13% with LVEF $\leq 40\%$), 763 had tetralogy of Fallot (1.5% with LVEF $\leq 40\%$), 120 had tricuspid atresia (5.5% with LVEF $\leq 40\%$), and 109 were ventricularly paced (5.8% with LVEF $\leq 40\%$). For a comprehensive list of all individual CHD lesions analyzed at CHOP, see Supplementary Table 1.

The Toronto General Hospital (TGH) external cohort was more representative of an adult CHD cohort; it comprised 629 ECG-CMR pairs (284 patients with repaired tetralogy of Fallot;

median age 34.8 [IQR 25.9, 44.4] years; 59% male), 23% with LVEF $\leq 50\%$ and 12% with LVEF $\leq 40\%$.

Robustness to Simulated Real-Life Noise

We first evaluated model robustness to simulated real-life noise encountered in portable single-lead ECG devices.

Without added noise, the baseline and noise-trained models performed similarly for predicting LVEF $\leq 50\%$ (AUROC 0.86; $p=0.94$) and LVEF $\leq 40\%$ (AUROC 0.91–0.92; $p=0.61$) (Figure 2). However, under increasing noise levels, baseline model performance deteriorated substantially. At a SNR of 0.5, baseline model performance dropped to AUROC 0.52 (LVEF $\leq 50\%$) and 0.44 (LVEF $\leq 40\%$). In contrast, the noise-trained model maintained high accuracy at the same noise level (AUROC 0.86 and 0.91, respectively; $p<0.0001$ vs. baseline), demonstrating substantial robustness to simulated real-world noise (Figure 2).

Noisy AI-ECG Model Performance

The noisy AI-ECG model showed strong discrimination of mild (LVEF $\leq 50\%$) and moderate (LVEF $\leq 40\%$) dysfunction during internal (Figure 3; left) and external CHOP (Figure 3; right) testing. Performance was similar when using either the first, last, or a random ECG per patient (Figure 3), demonstrating consistent performance across the CHD lifespan. The model is well calibrated for the outcomes LVEF $\leq 50\%$ (Brier score 0.07) and LVEF $\leq 40\%$ (Brier score 0.03).

Performance metrics for low-risk and high-risk cutoffs across institutions are shown in Table 2. Across institutions, model performance at the low-risk threshold for LVEF $\leq 40\%$ yielded 91–92% sensitivity, identifying $\sim 75\%$ of ECGs as negative with an NPV $\geq 99.8\%$. At the high-risk threshold, specificity reached 97–98%, with $\sim 3\%$ of ECGs flagged as positive and PPV ranging from 12–16%. Patients deemed high-risk had a positive likelihood ratio (i.e., an increase

in post-test odds of LVSD) of between 15.7 (internal testing) and 27.3 (external testing), conferring a strong indication for echo confirmatory testing. In contrast, patients deemed low-risk had a negative likelihood ratio between 0.12 (BCH) and 0.09 (CHOP), decreasing post-test odds of having LVSD by factors of 8.2 and 10.8, respectively. Between 23.5% (internal testing) and 16.3% (external testing) of patients had an intermediate risk, with a corresponding clinically uninformative likelihood ratio of 1.9 and 2.3 for internal and external testing cohorts, respectively.

Subgroup Analysis

The noisy AI-ECG model performance across institutions is shown in Figure 4A for select lesion subgroups: cardiomyopathy, tetralogy of Fallot, single ventricle disease, and ventricular pacing. In general: 1) AUROC model performance was similar across institutions for each lesion; and 2) similar AUROCs were obtained across lesions, with the notable exception of a lower AUROC in ventricular paced patients (internal AUROC 0.64, external AUROC 0.61).

As shown in Figure 4B, 15% of cardiomyopathy patients were classified as high-risk, of whom 43% had LVEF $\leq 40\%$; 46% were classified as low-risk, with only 2% having LVEF $\leq 40\%$. For tetralogy of Fallot, 7% were high-risk (12% with LVEF $\leq 40\%$), and 48% were low-risk (with $<0.1\%$ having LVEF $\leq 40\%$). In tricuspid atresia, 17% were high-risk (34% with LVEF $\leq 40\%$) and 30% low-risk (0% with LVEF $\leq 40\%$). Among ventricularly paced patients, 37% were classified high-risk, with 17% having LVEF $\leq 40\%$. The model was better calibrated in cardiomyopathy (Brier score 0.095) and tetralogy of Fallot (Brier score 0.021) subgroups than tricuspid atresia (Brier score 0.17) and ventricular pacing (Brier score 0.11) subgroups.

Internal (BCH) and external (CHOP) model performance across a broader range of CHD lesions is shown in Figure 4C and Supplementary Table 2. AUROCs ranged from a low of 0.68 in

truncus arteriosus to a high of 0.93-0.96 in total anomalous pulmonary venous connection. As shown in Supplementary Table 3, high performance is achieved across all race subgroups.

Prognostic Value of the Noisy AI-ECG Model

The prognostic utility of the noise-trained model was assessed using a random ECG per patient (Figure 5). High-risk classifications were associated with a 7.9-fold increased hazard of all-cause mortality compared to low-risk (Figure 5A). Significant survival differences were observed in cardiomyopathy ($p<0.001$), tetralogy of Fallot ($p<0.001$), tricuspid atresia ($p=0.04$), and a trend in ventricular pacing ($p=0.07$) (Figure 5B).

Model Explainability

Saliency mapping and median waveform analysis were performed to generate hypotheses of ECG features driving LVSD model predictions (Figure 6). Reassuringly, the saliency maps for the baseline and noise-augmented ECGs were remarkably similar, suggesting the AI-model is learning to filter out the simulated real-world noise to focus on the true predictive features of the underlying signal.

Key predictive features included prolonged PR intervals, widened QRS complexes, and reduced T wave amplitudes. These features were consistent across the overall cohort and subgroups, with the QRS complex consistently emerging as the most salient feature.

DISCUSSION

Portable, scalable, and accessible AI-enabled smartwatch technology has shown promise as a cardiovascular risk stratification strategy in the general adult population.^{11,14,15} The challenges of lifelong care in CHD—inclusive of gaps in care,⁶ loss to follow-up,⁷ regional and socioeconomic disparities worldwide,⁸ and demands of interstage remote surveillance¹⁰—underscore the need for similar solutions tailored to this population. Herein, we present the first and most comprehensive (to our knowledge) noise-adapted single-lead ECG model for predicting LVSD in pediatric CHD patients. Trained on >130,000 ECG-echo pairs from ~50,000 patients, our model was robust to simulated real-life wearable noise with strong performance across a broad range of CHD lesions, races, ages, and healthcare systems. In addition, this model provided prognostic value, predicting future all-cause mortality in multiple CHD populations. These findings support the potential of AI-enabled wearables for CHD risk stratification in remote monitoring and telemedicine, with applications in expanding access, bridging care gaps, and prioritizing patients for further evaluation or intervention.

Ventricular dysfunction can arise throughout the lifespan in CHD, including the high-risk interstage period for single ventricle patients.¹⁶ Heart failure contributes to ~20% of adult CHD hospitalizations¹⁷ and 20–40% of mortality.¹⁸ The pathophysiology of myocardial dysfunction and heart failure in CHD is multifactorial, involving hemodynamic and electrophysiologic abnormalities such as volume and pressure overload, surgical sequelae, arrhythmias, chronic pacing, and fibrosis.

Access to care and gaps in care are major roadblocks to early detection of LVSD and heart failure in CHD, especially in socioeconomically disadvantaged populations. Nearly half of the United States population lives >1 hour away from an adult CHD center, and 5% live >4

hours away.^{19,20} In Africa, an estimated 90% of all children with CHD do not have access to appropriate medical care.²¹ In Rwanda, there are two Rwandan pediatric cardiologists and one adult cardiothoracic surgeon (with no pediatric cardiac surgeon).²² In Zimbabwe, prior to 2021, only a single pediatric cardiologist served a population of ~15 million.²³ Clearly, gaps in care in CHD are increasing, with barriers disproportionately affecting socioeconomically disadvantaged populations.^{19,20} Furthermore, even in high-resource settings, the transition period from pediatric to adult CHD care is particularly vulnerable: in a multi-center study across 12 adult CHD centers, 42% of patients experienced gaps >3 years, and 8% had gaps exceeding a decade.²⁴ Remote home monitoring—especially for interstage patients—may help partially mitigate these disparities and gaps in care. In several recent studies, indices of socioeconomic status were associated with pre-operative risk factors,²⁵ but not post-operative interstage outcomes,^{25,26} in single ventricle patients (whereby remote patient monitoring is hypothesized to be contributing to similar post-operative interstage outcomes across socioeconomic groups).^{27,28} AI-enabled wearables may help facilitate larger-scale remote monitoring efforts in CHD.

Portable ECG recording wearables have several advantages compared to the standard 12-lead ECG machine: they are less expensive (≥ 10 -fold cheaper), more accessible (i.e., outside of the clinical setting), and a quicker, point-of-care modality. AI-enabled wearables have emerged as promising tools for cardiovascular surveillance in the general adult population. In 2022, Attia et al. developed and prospectively validated an AI-enabled consumer smartwatch ECG to detect LVEF $\leq 40\%$.¹⁴ Similar models have been subsequently developed to predict LVSD,¹⁵ risk of heart failure,¹¹ structural heart disease,²⁹ and hyperkalemia.³⁰ These advances support the feasibility of remote surveillance in CHD using similar technology, potentially reducing disparities in access to care.

While pediatric ECG wearables are of great interest,^{31,32} their clinical use remains limited. To date, studies have simply confirmed that devices like the Apple Watch^{33,34} and AliveCor Kardia³⁵ can produce accurate single-lead ECGs across pediatric age groups. The FDA has also approved an electronic stethoscope-based single-lead ECG device for patients ≥ 10 kg. Each device may provide a platform to deploy the algorithm presented herein.

The prospect of developing scalable and accessible technology for early detection of markers closely linked to heart failure (such as LVSD) in CHD motivated our efforts herein. Prediction of LVSD offers several benefits: 1) it reflects overall ventricular health; 2) it carries prognostic value³; 3) it may prompt initiation of GDMT.³ Clinically, we envision our tool may: 1) tackle barriers to care; 2) enable improved telemedicine efforts and reduced care gaps; 3) lead to earlier LVSD detection (which may lead to closer monitoring and/or earlier initiation of GDMT); and 4) promote longitudinal risk tracking across the CHD lifespan.

As highlighted above, there are profound disparities in pediatric cardiology care worldwide, especially in low- and middle-income countries. Importantly, as shown in Supplementary Table 3, model performance remains high across racial groups. In regions with limited pediatric cardiologist (and sonographer) availability, this tool may help screen for LVSD by a community healthcare provider which may help prioritize referrals to distant healthcare systems. Portable devices capable of recording single-lead ECGs could be deployed in underserved communities within a school or village. In such regions, cardiovascular surveillance of youth may be theoretically supported by a school nurse or primary care physician using a portable single-lead ECG device. For patients with other barriers to care (e.g., transportation issues, childcare or work-related issues), this technology may help risk stratify timing of in-person follow-up. AI-enabled portable devices could also facilitate telemedicine visits for

patients with CHD in underserved areas. For example, a community primary care provider could theoretically collect vital signs, digital recordings of cardiopulmonary auscultation, and AI-enabled prediction of ventricular dysfunction with an electronic stethoscope–based single-lead ECG. This information could then be transmitted to a pediatric cardiologist for triaging. For interstage home monitoring programs, AI-enabled smartwatches could predict LVSD, which may trigger closer follow-up, arch evaluation, and potentially earlier initiation of GDMT. Portable devices could support community-wide health promotion strategies in adults with CHD—especially for those unlikely to access healthcare settings—and facilitate earlier detection of LVSD. Remote portable device “function checks” at home or at the primary care physician office between cardiologist appointments may also lead to earlier detection of LVSD. Finally, for specific lesions like tetralogy of Fallot where the model is well calibrated, longitudinal tracking of LV systolic function across the lifespan may be feasible and beneficial for prognostication^{36,37}—especially during periods of high risk for care interruption.

Several limitations warrant discussion. First, clinical ECG lead I signals may differ from those captured by portable devices. While our simulated real-life noise-augmentation strategy supports model robustness, prospective validation on true wearable-derived ECGs is essential. Second, selection bias may exist due to reliance on documented LVEF, thus excluding cases where LVEF is infrequently reported (e.g., HLHS, L-loop TGA). In our previous work, we demonstrated performance was similar even when using gold standard ECG-CMR pairs on such patients.⁵ To this end, our model is limited to assessing left, not right, ventricular health. Future work may include model development to predict right ventricular health. Third, performance was lower for select cases (e.g., ventricular pacing). To address this, future directions include compilation of a larger heterogeneous dataset via multicenter collaboration and use of federated

learning.³⁸ Fourth, external validation was limited to tertiary centers, underscoring the need to assess generalizability across care settings. Finally, reliance on institutional coding systems introduces risk of lesion and mortality misclassification. Future work includes assessment of other rare lesions (e.g., coronary anomalies).

In conclusion, our noise-adapted, single-lead AI-ECG model reliably predicts LVSD across diverse CHD populations, demonstrating resilience to simulated real-world noise and consistent performance across institutions. As single-lead ECGs become more accessible through wearable and portable technologies, future studies should assess whether this digital biomarker can enhance LVSD risk stratification and CHD care delivery in underserved communities.

METHODS

This study is reported in accordance with the TRIPOD+AI 2024 guidelines.³⁹ Use of BCH, TGH, and CHOP data were approved by their respective Institutional Review Boards.

Internal Study Population and Patient Assignment

Internal patient data was derived from BCH from the earliest digitized ECG in 1987 until January 2023, inclusive of any patient with ≥ 1 echo with a recorded LVEF. Only transthoracic echos were included; fetal echos were excluded. Each echo was paired to the closest ECG, whereby only ECG-echo pairs ≤ 2 days apart were included. For quality control, ECGs shorter than 10 seconds or with missing lead data were excluded. The resulting ECG-echo pairs comprised the main cohort and were split at the patient level (to avoid data leakage) into training (80%) and test (20%) sets.

Our cohort included patients with and without CHD, given that: 1) it is plausible that there are shared electrophysiological signatures of LV systolic dysfunction among CHD and non-CHD ECGs; 2) it enhances cohort diversity, which may lead to more robust detection of the fundamental electrophysiologic features of systolic dysfunction; and 3) it may help avoid overfitting, as limiting training to CHD would shrink the sample size and reduce morphological diversity, risking overfitting to specific CHD lesions.

External Study Populations

External validation was performed at CHOP and TGH. At CHOP, all ECG-echo pairs between 2000-2023 meeting the same inclusion and exclusion criteria described above were used. At TGH, inclusion criteria consisted of: 1) repaired TOF patients from the INDICATOR cohort⁴⁰; 2) ≥ 1 cardiac magnetic resonance (CMR) imaging study with a recorded LVEF; and 3) at least one ECG-CMR pair ≤ 30 days apart without an intermediate pulmonary valve replacement. ECGs

failing quality control described above were excluded. The remaining ECG-CMR pairs comprised the external TGH cohort.

Data Collection

Raw ECG waveforms were extracted from the MUSE ECG management system (GE Healthcare, Chicago, IL) at BCH, CHOP, and TGH.

LVEF values were obtained from imaging reports, where the LV always corresponds to the morphologic LV. At BCH and CHOP, LVEF was obtained from echo reports using the bullet method at BCH and the Simpson's disc method at CHOP. At TGH, LVEF was obtained from CMR reports.

CHD Diagnostic Coding

CHD diagnoses were determined from echo reports using the Fyler coding system at BCH and ICD-9/10 codes at CHOP. At BCH and CHOP, a patient was classified as having a specific lesion if any of their ECG-echo pairs contained the corresponding diagnostic code.

BCH uses the Fyler coding system, a detailed and decades-established anatomic classification system specifically designed for CHD. For every echo performed, cardiac imagers assign Fyler codes that capture both major and minor structural lesions with high granularity. These codes were manually validated during the early stages of our prior AI-ECG work, and their accuracy has since been supported by ongoing efforts, including internal and external validation of AI-ECG models for detecting critical CHD in infants.⁴¹

At CHOP, echo reports were identified as the most reliable source of information which could be queried for cardiac anatomy classification within the medical record. A multidisciplinary working group created a system which maps unique ICD-9 and ICD-10

diagnostic codes to specific and clinically appropriate disease categories (e.g., Tetralogy of Fallot, VSD, hypoplastic left heart syndrome, etc.).

Using these institution-specific systems, cardiomyopathy was identified from echo-associated diagnoses by Fyler codes at BCH and ICD-9/10 codes at CHOP. Cardiomyopathy included dilated, hypertrophic, restrictive, arrhythmogenic, non-compaction, metabolic/mitochondrial, and unspecified cardiomyopathy. At both institutions, imagers would not classify a patent foramen ovale as an atrial septal defect unless there is a strong clinical suspicion of a true defect. At both institutions, ventricularly paced patients were identified based on ECG diagnoses of “dual chamber pacing” or “ventricular pacing”.

Data Preprocessing and Conversion to Noisy Single-Lead ECGs

Two separate single-lead ECGs were generated in this study: a baseline single-lead ECG and a noisy single-lead ECG.

To generate the baseline single-lead ECG, lead I was isolated from each 12-lead ECG. Lead I was selected because it most closely corresponds to the recorded ECG lead that is obtained with a portable device. This ECG was then resampled to 500 Hz and trimmed to 4096 samples (~8 seconds) to facilitate integration with a convolutional neural network. This resultant single-lead ECG without additional noise was defined as the baseline single-lead ECG.

Noisy single-lead ECGs were derived from 12-lead ECGs using previously established methods.^{11,15} Wearable ECG data are exposed to several sources of noise, including but not limited to poor electrode contact, movement and muscle contraction artifact, and external electrical interference.^{11,15} To simulate real-life noise, the baseline single-lead ECG was noise augmented using random gaussian noise generated at four frequency ranges known to correlate with sources of portable/wearable ECG artifact: 3-12 Hz (motion artifact during tremors), 12-50

Hz (lower-frequency muscle activation artifact), 50-100 Hz (electrode motion noise), and 100-150 Hz (higher-frequency muscle activation artifact).^{11,15} Powerline interference noise is further captured within the 50-100 Hz and 100-150 Hz frequency ranges.⁴²

Noisy single-lead ECGs were generated by: 1) isolating lead I from each 12-lead ECG; 2) resampling to 500 Hz; 3) exposing each 10-second lead I sequence randomly to one of the four frequency-banded gaussian noises at a $\text{Log}_2(\text{signal-to-noise ratio})$ randomly between -1 and 1 (corresponding to a signal-to-noise ratio between 0.5 and 2); and 4) trimming to a random 4096 sample interval (~8 seconds) within the 10-second sequence. This preprocessing was applied uniformly across BCH, CHOP, and TGH datasets.

Definition of Primary Outcomes

The primary outcome was $\text{LVEF} \leq 40\%$ (corresponding to moderate or worse dysfunction). The secondary outcome was $\text{LVEF} \leq 50\%$ (corresponding to mild or worse dysfunction). As secondary analyses for prognostic value, we also evaluated time-to-mortality (see “Time-to-Event Analysis” below).

Convolutional Neural Network Model Development

Two AI-ECG models were generated on the training set: 1) baseline model (trained on the baseline single-lead ECG); and 2) noisy model (trained on the noisy single-lead ECG). Two noise-added ECGs were generated from each baseline ECG, effectively doubling the noisy model training set size.

The training dataset was split (95% training, 5% validation) for hyperparameter tuning. 1 x 4096 ECG samples were used as inputs to a convolutional neural network with a residual block architecture that has been adapted for unidimensional signals as previously described (Supplementary Figure 1).^{5,43}

Hyperparameters were tuned via a grid search (kernel size [3, 9, 17], batch size [8, 32, 64], and initial learning rate [0.01, 0.001, 0.0001]) using Adam optimization and early stopping (patience of 5 epochs, maximum 150 epochs). The models with the lowest loss in the validation set were selected (for both models: kernel size 17, batch size 32, learning rate 0.001).

Performance Evaluation and Statistical Analysis

Multiple ECG-echo pairs per patient were allowed in the training cohort. In the overall test cohorts, one randomly selected noisy ECG-echo pair per patient was used for evaluation. To evaluate temporal robustness across the lifespan, performance was also assessed on the first or last available noisy ECG-echo pair per patient.

Given the imbalanced dataset, metrics included area under the receiver operating curve (AUROC), area under the precision-recall curve (AUPRC), positive predictive value (PPV), negative predictive value (NPV), positive likelihood ratio, negative likelihood ratio, sensitivity, and specificity. Thresholds were selected to achieve 95% sensitivity (defined as low-risk cutoff) and 95% specificity (defined as high-risk cutoff) on the validation set. Patients with predictions in between the low- and high-risk cutoff were deemed intermediate risk. Confidence intervals were calculated via resampling of 1,000 bootstraps.

To assess model calibration, the Brier score was obtained (where 0 is perfect calibration, 1 is total miscalibration).

Subgroup Analysis

Model performance was explored across a wide spectrum of CHD lesion subgroups at BCH and CHOP. Select CHD subgroup populations were deemed of special interest for this technology: patients with cardiomyopathy, tetralogy of Fallot, single ventricle disease, and ventricular pacing. Given that many lesions are on a spectrum for surgical strategies (including hypoplastic

left heart syndrome^{44,45}), we opted to solely use tricuspid atresia to represent the single ventricle disease category, as: 1) single ventricle palliation is always pursued with this lesion; 2) the morphological LV is the systemic ventricle; and 3) LVEF is more frequently reported than other single ventricle lesions. To compensate for small sample sizes, all eligible ECG-echo pairs ≤ 2 days apart were used in subgroup analyses.

Prognostic Analysis

All-cause mortality was captured within the BCH Fyler coding system. Patients were stratified into low-, intermediate-, or high-risk groups based on a single random noisy single-lead ECG per patient. Time-to-event analysis for all-cause mortality was then conducted using Cox proportional hazards models adjusted for age. Statistical comparison between groups based on log-rank testing were also reported. Patients who did not experience all-cause mortality were censored at time last known alive.

Model Interpretability

Model explainability was conducted via median waveform analysis and saliency mapping as previously described.^{43,46-48} Median waveform analysis helps visualize examples of high-risk and low-risk ECGs. Saliency mapping helps identify which features of the ECG input contribute to model prediction.

Representative waveforms and saliency maps were generated from $n=100$ samples (overall cohort) and $n=25$ (lesion-specific subgroups), consistent with prior work.^{5,43,46}

DATA AVAILABILITY

The datasets generated and/or analyzed during the current study are not publicly available due to patient health information but are available from the corresponding author on reasonable request.

Requests for BCH data and related materials will be internally reviewed to clarify if the request is subject to confidentiality constraints. Shareable data and materials will be released under a material transfer agreement for non-commercial research purposes.

CODE AVAILABILITY

Models were developed using Keras with a TensorFlow backend in Python 3.9. Pre- and post-processing were performed using Python 3.9 and R 4.0. Training was conducted on institutional GPUs. Requests for programming code and related materials will be internally reviewed to clarify if the request is subject to intellectual property. Shareable data and materials will be released under a material transfer agreement for non-commercial research purposes.

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AUTHOR CONTRIBUTIONS

JM and IA contributed to data collection and data management. JM, IA, and JT contributed to study coordination. JM, IA, and WL contributed to data analysis. All authors contributed to the

data interpretation and writing of the manuscript. JM, IA, and JT contributed to the study design.

All authors contributed to the critical review and editing of the manuscript.

COMPETING INTERESTS

None.

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FIGURES

Figure 1: Study Overview. To evaluate whether artificial intelligence (AI)-enabled wearables could predict left ventricular ejection fraction (LVEF) in congenital heart disease (CHD), we adapted standard 12-lead electrocardiograms (ECGs) into simulated real-life noisy single lead ECGs. The model was trained and validated at Boston Children's Hospital (BCH), with multinational external validation at Children's Hospital of Philadelphia (CHOP) and Toronto General Hospital (TGH).

Figure 2: Noisy Single-Lead Model is Robust to Simulated Real-Life Noise. (A) Representative single-lead ECG with no noise added, as well as increasing noise (i.e., signal-to-noise ratios [SNRs] of 2 to 0.5). Baseline (black) and noisy (blue) model performance via area under the receiver operating curves (AUROC) for the outcome left ventricular ejection fraction (LVEF) of **(B)** $\leq 50\%$ and **(C)** $\leq 40\%$ when applying increasing noise (noise increasing from left to right panels). 95% confidence intervals are shown using bootstrapping. P-values obtained via the DeLong test.

Figure 3: Internal Testing and External Validation of the Noisy Single-Lead AI-ECG Model to Predict Left Ventricular Systolic Dysfunction. Performance of the noisy single-lead artificial intelligence-enabled ECG algorithm in the internal Boston Children's Hospital (BCH; left) and external Children's Hospital of Philadelphia (CHOP; right) cohorts using receiver operating (AUROC) and precision recall (AUPRC) curves for the following outcomes: left ventricular ejection fraction (LVEF) $\leq 50\%$ and LVEF $\leq 40\%$. Model performance assessed on a single ECG-echo pair per patient: random (blue), first (orange), and last (green). AUROC and AUPRC metric values for each model and outcome are inset. Dotted line represents chance 95% confidence intervals are shown using bootstrapping.

Figure 4: Model Performance Across Congenital Heart Disease Lesion Subgroups. (A) Internal testing at Boston Children's Hospital (BCH; blue) and external validation at Children's Hospital of Philadelphia (CHOP; grey) or Toronto General Hospital (TGH; purple) performance using area under the receiver operating (AUROC) and precision recall (AUPRC) curves for select lesions: cardiomyopathy, tetralogy of Fallot, tricuspid atresia, and ventricular pacing. **(B)** Proportion of low- (green), intermediate- (yellow), high-risk (red) patients with left ventricular ejection fraction (LVEF) $\leq 40\%$. **(C)** Forest plot showing internal testing (black) and CHOP external validation (red) AUROC and AUPRC performance when stratifying by a wide range of congenital heart disease lesions (top-to-bottom: sorted from highest to lowest internal AUROC) for the outcome left ventricular ejection fraction (LVEF) $\leq 40\%$. 95% confidence intervals are shown using bootstrapping. **Abbreviations:** atrial septal defect (ASD); complete atrioventricular canal (CAVC); coarctation of the aorta (CoA); double outlet right ventricular (DORV); transposition of the great arteries (TGA); hypoplastic left heart syndrome (HLHS); total anomalous pulmonary venous connection (TAPVC); ventricular septal defect (VSD).

Figure 5: Prognostic Value of the Noisy Single-Lead AI-ECG Model. Kaplan-Meier curve survival analysis when stratifying patients as low-, intermediate-, or high-risk based on AI-ECG left ventricular systolic dysfunction probabilities for the **(A)** overall cohort and **(B)** select subgroups: cardiomyopathy, tetralogy of Fallot, tricuspid atresia, and ventricular pacing. Color-

coded hazard ratio (HR) with 95% confidence interval (CI) inset below the overall cohort (obtained via Cox regression analysis when adjusting for age, with the low-risk group as reference). Log-rank p-value inset.

Figure 6: Explainability of AI-ECG Predictions. Visualization of median waveforms generated in lead I ECGs from the highest (red) and lowest (green) AI-ECG predictions of the (A) overall test cohort with and without noise, as well as (B) cardiomyopathy, tricuspid atresia, tetralogy of Fallot, and ventricular pacing subgroups. Saliency mapping demarcates regions of the ECG waveform having greatest (dark blue) and least (light blue) influence on each outcome. Saliency was averaged over the highest predicted ECGs for left ventricular ejection fraction $\leq 40\%$.

TABLES

Table 1: Baseline Characteristics of Internal and External Cohorts

Table 2: Model Performance Metrics During Internal and External Testing

Table 1: Baseline Characteristics of Internal and External Cohorts				
	BCH (Training)	BCH (Testing)	CHOP	TGH
Overall				
Patients	52,670	17,556	42,984	284
Sex (male)	28,137 (53%)	9,425 (54%)	23,815 (55%)	168 (59%)
ECG-echo pairs	133,344	45,151	76,400	629*
Age at ECG (years)	10.7 (3.5, 16.8)	10.5 (3.5, 16.8)	10.8 (4.9, 15.0)	34.8 (25.9, 44.4)
LVEF $\leq$ 50%	9,091 (6.8%)	3,108 (6.9%)	2,888 (3.8%)	145 (23%)
LVEF $\leq$ 40%	3,571 (2.7%)	1,283 (2.8%)	1,313 (1.7%)	75 (12%)
Race/Ethnicity				
White (non-Hispanic)	30,887 (59%)	10,379 (59%)	—	—
Black (non-Hispanic)	3,010 (5.7%)	961 (5.5%)	—	—
Asian (non-Hispanic)	1,6781 (3.2%)	581 (3.3%)	—	—
Hispanic	4,808 (9.1%)	1,609 (9.2%)	—	—
Other	3,303 (6.3%)	1,145 (6.5%)	—	—
Unknown	8,991 (17%)	2,881 (16%)	—	—
Cardiomyopathy				
Patients	2,579	909	1,808	—
Sex (male)	1,516 (59%)	536 (59%)	1076 (59.5%)	—
ECG-echo pairs	19,810	6,781	5,158	—
Age at ECG (years)	14.2 (6.9, 19.1)	13.8 (6.9, 18.6)	9.7 (3.5, 14.6)	—
LVEF $\leq$ 50%	4,345 (22%)	1,564 (23%)	1,139 (22.1%)	—
LVEF $\leq$ 40%	2,212 (11%)	843 (12%)	659 (12.8%)	—
Tetralogy of Fallot				
Patients	2,027	697	763	284
Sex (male)	1,133 (56%)	388 (56%)	441 (58%)	168 (59%)
ECG-echo pairs	9,524	3,564	2094	629*
Age at ECG (years)	12.7 (4.4, 22.1)	11.6 (4.1, 21.3)	8.0 (3.8, 12.8)	34.8 (25.9, 44.4)
LVEF $\leq$ 50%	758 (8.0%)	258 (7.2%)	86 (4.1%)	145 (23%)
LVEF $\leq$ 40%	179 (1.9%)	78 (2.2%)	32 (1.5%)	75 (12%)
Tricuspid Atresia				
Patients	249	90	120	—

Sex (male)	135 (54%)	54 (60%)	73 (60.8%)	—
ECG-echo pairs	1084	391	271	—
Age at ECG (years)	11.2 (3.6, 19.3)	11.6 (4.7, 19.1)	6.2 (3.1, 11.8)	—
LVEF $\leq$ 50%	195 (18%)	105 (27%)	27 (10.0%)	—
LVEF $\leq$ 40%	62 (5.7%)	47 (12%)	15 (5.5%)	—
Ventricularly Paced				
Patients	698	253	109	—
Sex (male)	334 (48%)	126 (50%)	62 (56.9%)	—
ECG-echo pairs	2,817	1,060	206	—
Age at ECG (years)	13.9 (5.8, 23.7)	12.1 (4.9, 20.2)	9.7 (5.0, 13.7)	—
LVEF $\leq$ 50%	844 (30%)	332 (31%)	33 (16.0%)	—
LVEF $\leq$ 40%	373 (13%)	130 (12%)	12 (5.8%)	—

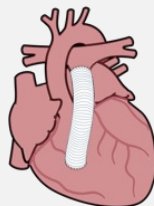
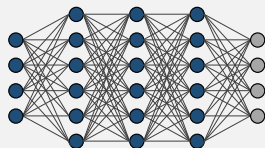
* Number of pairs in the TGH cohort represents ECG-cardiac magnetic resonance pairs.
Data presented as median (interquartile range).

Table 2: Model Performance Metrics During Internal and External Testing

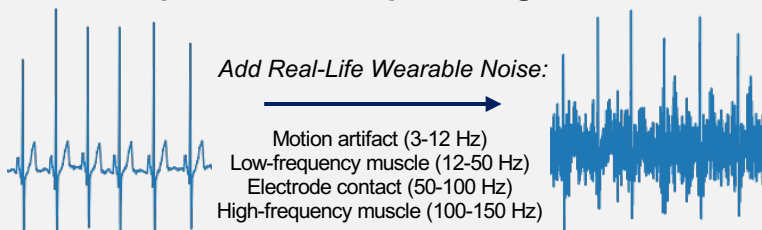
	BCH	CHOP
Low-Risk Cutoff		
<i>LVEF $\leq$ 50%</i>		
Sensitivity	0.95 [0.93-0.97]	0.88 [0.85-0.90]
Specificity	0.47 [0.46-0.48]	0.54 [0.54-0.55]
Negative Predictive Value (%)	99.6 [99.5-99.7]	99.7 [99.6-99.7]
Positive Predictive Value (%)	6.4 [6.3-6.6]	2.7 [2.6-2.8]
Predicted Negative (%)	45.6 [44.9-46.3]	53.5 [53.0-53.9]
<i>LVEF $\leq$ 40%</i>		
Sensitivity	0.91 [0.87-0.94]	0.92 [0.89-0.96]
Specificity	0.74 [0.73-0.74]	0.82 [0.81-0.82]
Negative Predictive Value (%)	99.8 [99.8-99.9]	100.0 [99.9-100.0]
Positive Predictive Value (%)	4.4 [4.2-4.6]	2.5 [2.3-2.5]
Predicted Negative (%)	72.7 [72.1-73.3]	81.4 [81.0-81.7]
High-Risk Cutoff		
<i>LVEF $\leq$ 50%</i>		
Sensitivity	0.34 [0.30-0.38]	0.43 [0.39-0.47]
Specificity	0.97 [0.97-0.97]	0.97 [0.97-0.97]
Negative Predictive Value (%)	97.5 [97.4-97.6]	99.2 [99.1-99.2]
Positive Predictive Value (%)	30.5 [27.7-33.4]	16.6 [15.1-18.1]
Predicted Positive (%)	4.1 [3.8-4.4]	3.7 [3.5-3.9]
<i>LVEF $\leq$ 40%</i>		
Sensitivity	0.47 [0.41-0.53]	0.55 [0.48-0.62]
Specificity	0.97 [0.97-0.97]	0.98 [0.98-0.98]
Negative Predictive Value (%)	99.3 [99.2-99.4]	99.8 [99.7-99.8]
Positive Predictive Value (%)	16.2 [14.2-18.4]	11.9 [10.3-13.4]
Predicted Positive (%)	3.8 [3.5-4.0]	2.3 [2.2-2.4]

ARTICLE IN PRESS

Can AI-enabled wearables predict LVEF in CHD?



CNN Input: Noise-Adapted Single-Lead ECG



Multi-Center External Validation

Training: BCH



52,670 patients
133,344 pairs
6.8% LVEF $\leq$ 50%
2.7% LVEF $\leq$ 40%

Internal
Testing: BCH



17,556 patients
45,151 pairs
6.9% LVEF $\leq$ 50%
2.8% LVEF $\leq$ 40%

External
Validation: CHOP

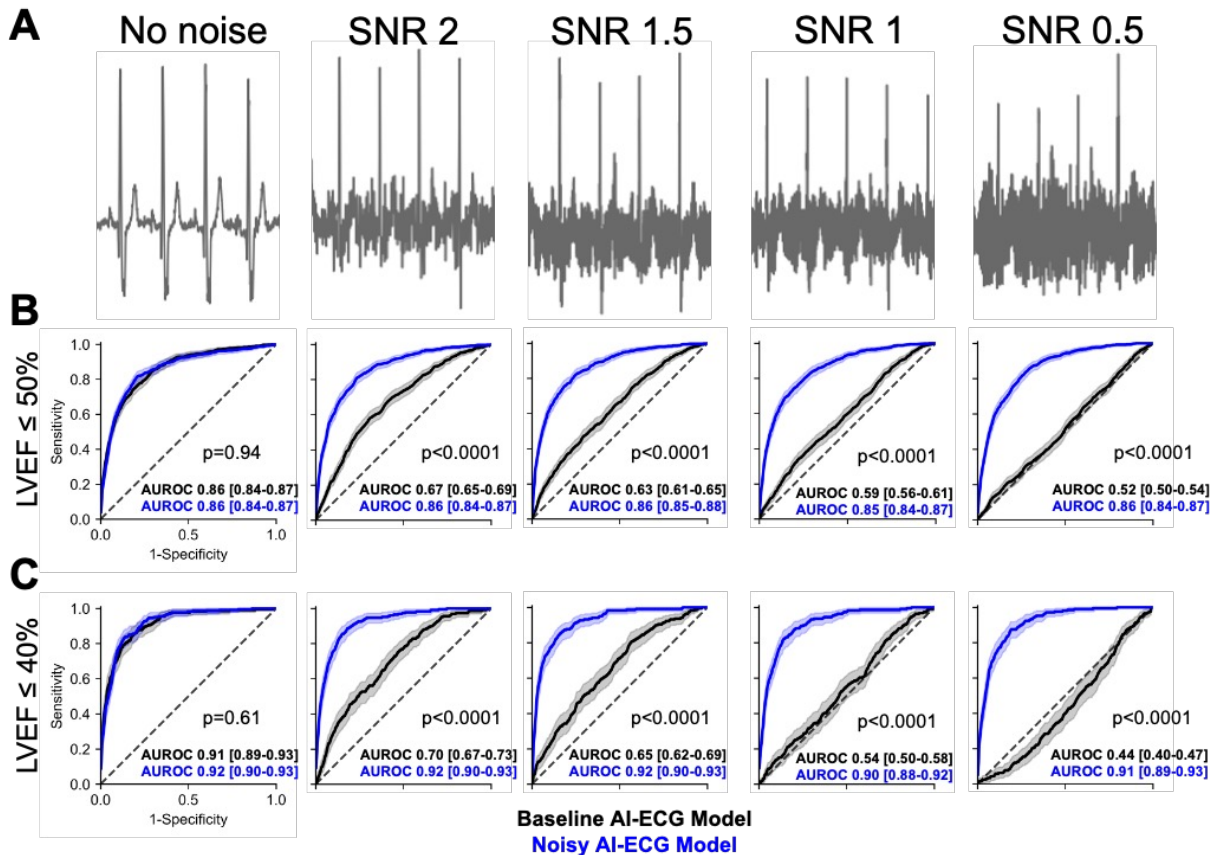


42,984 patients
76,400 pairs
3.8% LVEF $\leq$ 50%
1.7% LVEF $\leq$ 40%

External
Validation: TGH



284 patients
629 pairs
23% LVEF $\leq$ 50%
12% LVEF $\leq$ 40%



BCH

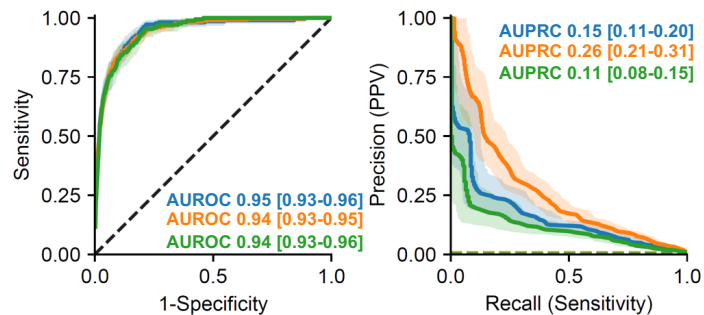
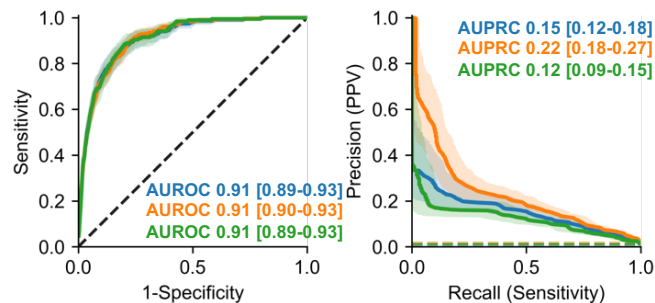
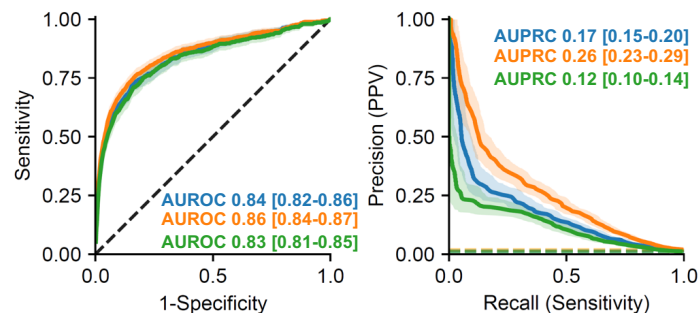
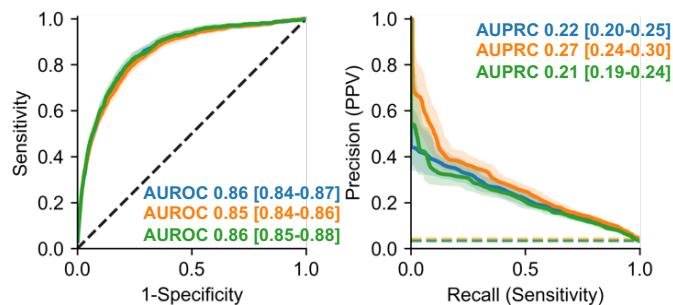


Random ECG

Last ECG

First ECG

CHOP

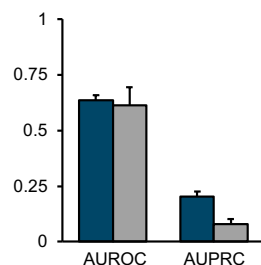
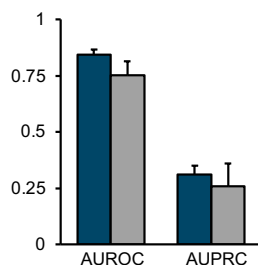
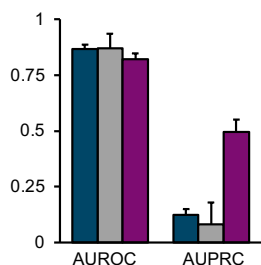
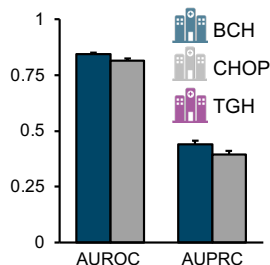
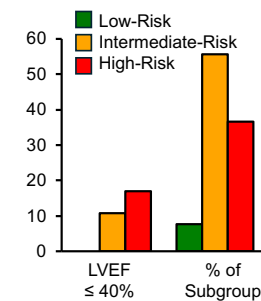
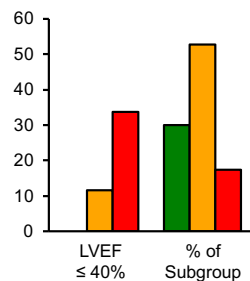
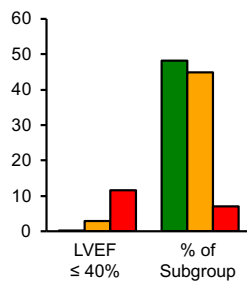
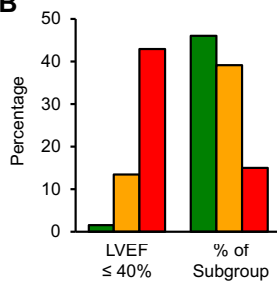
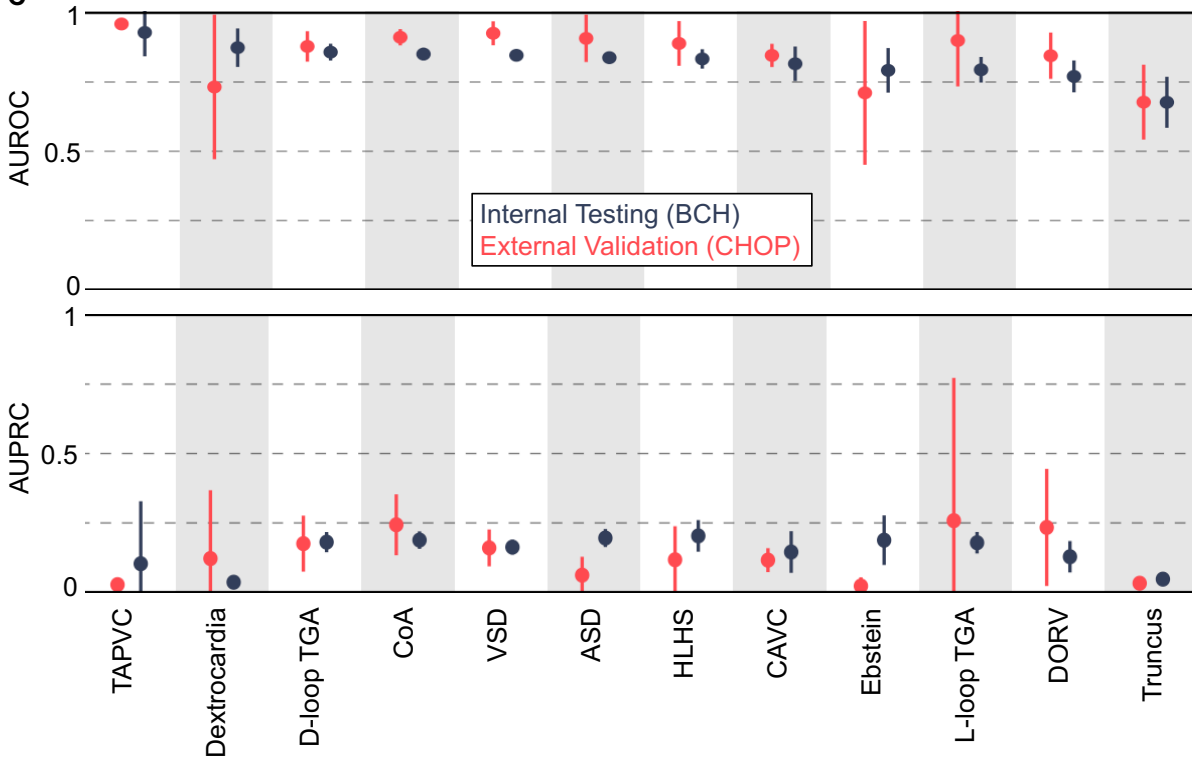


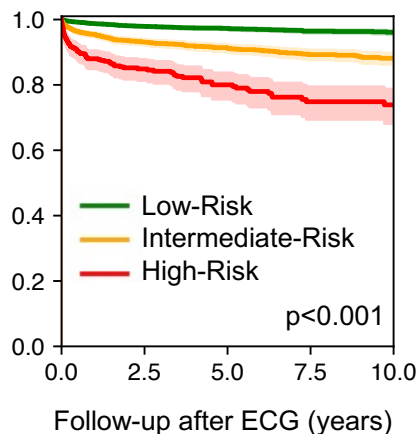
Cardiomyopathy

Tetralogy of Fallot

Tricuspid Atresia

Ventricular Pacing

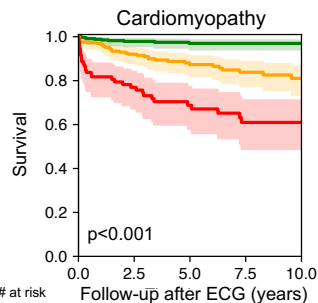
A**B****C**

A**Overall Cohort**

at risk

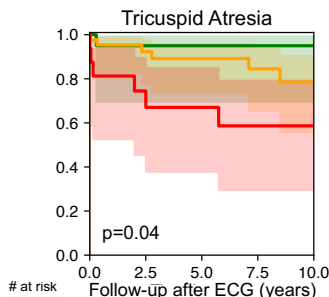
—	12701	2144	694
—	4185	898	332
—	670	167	77

HR 3.0 [95% CI, 2.5-3.7], $p < 0.001$
HR 7.9 [95% CI, 6.1-10.1], $p < 0.001$

B

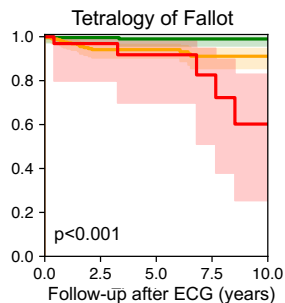
at risk

—	460	188	90
—	328	117	48
—	121	43	17



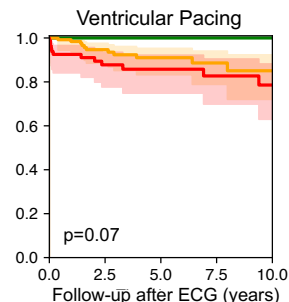
at risk

—	26	10	6
—	48	26	10
—	16	9	6



at risk

—	307	123	48
—	348	127	52
—	42	13	3

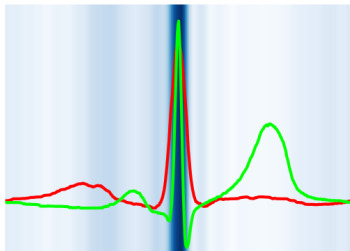


at risk

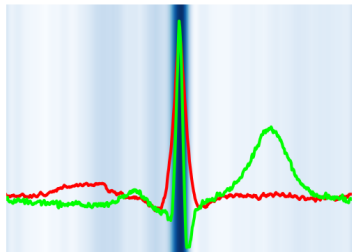
—	28	14	6
—	139	54	21
—	86	36	19

A

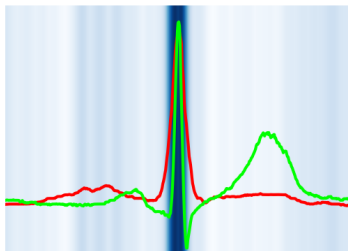
No noise



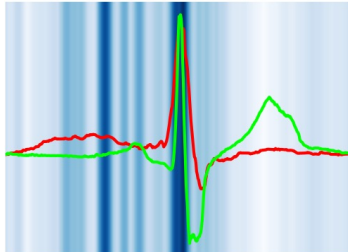
Noisy

**B**

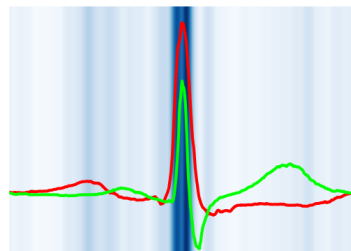
Cardiomyopathy



Tetralogy of Fallot



Tricuspid Atresia



Ventricular Pacing

